

Data Quality and Validation in ETL

Q1: Define Data Quality in the context of ETL pipelines. Why is it more than just data cleaning?

Ans.

Data Quality in ETL means ensuring that data is accurate, complete, consistent, and reliable before it is loaded into the target system.

It is more than just data cleaning because cleaning only fixes errors like nulls and duplicates. Data quality also covers validation against business rules, checking referential integrity, ensuring correct formats, and maintaining consistency across multiple sources.

Q2: Explain why poor data quality leads to misleading dashboards and incorrect decisions.

Ans.

Dashboards and reports are built on top of data. If the data has errors, duplicates, or missing values, the numbers shown on dashboards will be wrong. For example, duplicate transactions will inflate total sales figures. Business teams then make decisions based on these wrong numbers which leads to incorrect strategies and financial losses.

Q3: What is duplicate data? Explain three causes in ETL pipelines.

Ans.

Duplicate data means the same record appears more than once in the dataset.

Three causes in ETL pipelines:

1. Multiple source systems - Same data coming from two different systems gets loaded twice.
2. ETL job running multiple times - If the pipeline runs again without checking, same records get inserted again.
3. No primary key check - When there is no unique key validation, duplicate rows enter the target table easily.

Q4: Differentiate between exact, partial, and fuzzy duplicates.

Ans.

Exact Duplicates:

All column values are completely identical. Example - Same Txn_ID, Customer_ID, Amount repeated twice.

Partial Duplicates:

Most columns match but one or two columns are different. Example - Same customer and product but different date.

Fuzzy Duplicates:

Records that are similar but not exactly same due to typos or formatting. Example - Rahul Mehta and Rahul Mehta. are treated as different but are actually the same person.

Q5: Why should data validation be performed during transformation rather than after loading?

Ans.

If validation is done after loading, bad data already enters the target database. Removing or fixing it later is costly and time consuming. Validating during transformation catches errors early, prevents bad data from loading, and keeps the target system clean and reliable.

Q6: Explain how business rules help in validating data accuracy. Give an example.

Ans.

Business rules are conditions that data must satisfy based on real world logic. They help catch data that is technically valid but logically wrong.

Example - A business rule can say Quantity must not be Null and Txn_Amount must be greater than 0. In the dataset, row 205 has Quantity as Null and row 206 has Txn_Amount as Null. These rows would be flagged or rejected during validation based on these rules.

Q7: Write an SQL query to list all duplicate keys and their counts using the business key (Customer_ID + Product_ID + Txn_Date + Txn_Amount).

Ans.

```
SELECT Customer_ID, Product_ID, Txn_Date, Txn_Amount,  
       COUNT(*) AS duplicate_count  
FROM Sales_Transactions  
GROUP BY Customer_ID, Product_ID, Txn_Date, Txn_Amount  
HAVING COUNT(*) > 1;
```

Result from given dataset:

C101, P11, 2025-12-01, 4000 appears 3 times (Txn_ID 201, 203, 208) - these are exact duplicates.

Q8: Enforcing Referential Integrity - Identify Customer_ID values in Sales_Transactions that violate referential integrity with Customers_Master.

Ans.

Customers_Master has only C101, C102, C103, C104. Sales_Transactions has C105 and C106 which do not exist in Customers_Master. These violate referential integrity.

SQL Query to detect violations:

```
SELECT DISTINCT s.Customer_ID  
FROM Sales_Transactions s  
LEFT JOIN Customers_Master c  
ON s.Customer_ID = c.CustomerID  
WHERE c.CustomerID IS NULL;
```

Violating Customer IDs:

C105 (N/A - Pune) and C106 (Amit Verma - Pune) are not present in Customers_Master and violate referential integrity.